Chapter 4 Real function

Tags	Courses

Introduction

> session: 2/12/2024

Let $D \subseteq \mathbb{R}$. Often, we need to associate to an element $x \in \mathbb{R}$ a value, denoted by $f(x)$. This means that we define a **function** (also called an **application** or **map**).

1. Definition

Let $D \subseteq \mathbb{R}$. A function $f : D \to \mathbb{R}$ is a rule that assigns to every $x \in D$ a unique element $f(x) \in \mathbb{R}$.

- The **range** of f is:

$$\text{range}(f) = \{y \in \mathbb{R} \mid \exists x \in D : y = f(x)\}.$$

We write: $x \to f(x)$.

- D is called the **domain of definition** of f.
- The set $\{(x, f(x)) \mid x \in D\}$ is called the **graph of** f, usually denoted by $f(D)$.

> **Remark**
>
> If $f : D \to \mathbb{R}$ satisfies $f : D \to f(D)$, then it is called **surjective**.

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2. Terminology

- $f(x)$ is called the **image** of x by f.

- x is called the **independent variable** or **argument**.
- $f(x)$ or y (where $f(x) = y$) is called the **dependent variable**.

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3. Property

- The **composition** of functions f and g is defined as follows:

  $$f \circ g(x) = f(g(x)) \quad \text{and} \quad g \circ f(x) = g(f(x)),$$
  provided:

  $$\text{range}(g) \subseteq \text{domain}(f) \quad (\text{or} \quad \text{range}(f) \subseteq \text{domain}(g)).$$
- Domains of the compositions are:

  $$\text{domain}(f \circ g) = \text{domain}(g), \quad \text{domain}(g \circ f) = \text{domain}(f).$$

1. **Addition of Functions**

   For all $x \in D$:

   $(f + g)(x) = f(x) + g(x).$
2. **Multiplication of Functions**

   For all $x \in D$:

   $(f \cdot g)(x) = f(x) \cdot g(x).$
3. **Scaling a Function**

   For all $\lambda \in \mathbb{R}$ and $x \in D$:

   $(\lambda g)(x) = \lambda \cdot g(x).$
4. **Reciprocal of a Function**

   If $f(x) \neq 0 \quad \forall x \in D$:

   $\left(\frac{1}{f}\right)(x) = \frac{1}{f(x)}.$

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4. Equality of Functions

Let f and g be two real functions. We say that $f = g$ if and only if:

1. $\text{domain}(f) = \text{domain}(g),$
2. $f(x) = g(x) \quad \text{for all} \quad x \in \text{domain}(f).$

In general:

$$(f \circ g) \neq (g \circ f).$$

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5. Graph of a Function

Let f be a real function. The **graph of** f, denoted by $|f$ or $\text{graph}(f)$, is the set:

$$\text{graph}(f) = \{(x, f(x)) \mid x \in \text{domain}(f)\}.$$

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6. Even and Odd Functions

1. The domain of f, D_f, is **symmetric**, meaning:

$$\text{If } x \in D_f, \text{ then } -x \in D_f.$$

- **Even function:** $f(-x) = f(x) \quad \text{for all} \quad x \in D_f.$
- **Odd function:** $f(-x) = -f(x) \quad \text{for all} \quad x \in D_f.$

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7. Periodic Functions

A function $f : D \to \mathbb{R}$ is called **periodic** if there exists $T > 0$ such that:

1. $x + T \in D \quad \text{for all} \quad x \in D,$
2. $f(x + T) = f(x) \quad \text{for all} \quad x \in D.$

If f is T-periodic, then for all $x \in D$:

$$f(x + nT) = f(x), \quad n \in \mathbb{Z}.$$

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8. Bounded Functions

Definition 4.9

Let $f : D \to \mathbb{R}$ and let $f(D)$ be the set of all values of f.

- We say that function f is **bounded above** on its domain D if $f(D)$ is bounded above, i.e.,

  $\exists M \in \mathbb{R}, \forall x \in D : f(x) \leq M$

- We say that function f is **bounded below** on its domain D if $f(D)$ is bounded below, i.e.,

  $\exists m \in \mathbb{R}, \forall x \in D : f(x) \geq m$
- We say that function f is **bounded** on its domain D if $f(D)$ is bounded, i.e.,

  $\exists A \in \mathbb{R}, \forall x \in D : |f(x)| \leq A$

> ### Remark 4.3
>
> If f is bounded on D, then $f(D)$ admits a supremum M and an infimum m. We denote:
>
> $M = \sup_{x \in D} f(x), \quad m = \inf_{x \in D} f(x)$
>
> We have:
>
> $\sup_{x \in D} f(x) = M < +\infty \iff \begin{cases} \forall x \in D, f(x) \leq M \\ \forall \epsilon > 0, \exists x_0 \in D : f(x_0) > M - \epsilon \end{cases}$
>
> $\inf_{x \in D} f(x) = m > -\infty \iff \begin{cases} \forall x \in D, f(x) \geq m \\ \forall \epsilon > 0, \exists x_0 \in D : f(x_0) < m + \epsilon \end{cases}$

9. Monotone Functions

Definition 4.10

Consider $f : D \subset \mathbb{R} \to \mathbb{R}$, and set $M \subset D$.

1. f is **nondecreasing** on M if:

   $\forall x_1, x_2 \in M,\, x_1 \leq x_2 \implies f(x_1) \leq f(x_2)$
2. f is **nonincreasing** on M if:

   $\forall x_1, x_2 \in M,\, x_1 \leq x_2 \implies f(x_1) \geq f(x_2)$
3. f is **increasing** on M if:

   $\forall x_1, x_2 \in M,\, x_1 < x_2 \implies f(x_1) < f(x_2)$

4. f is **decreasing** on M if:

$$\forall x_1, x_2 \in M,\, x_1 < x_2 \implies f(x_1) > f(x_2)$$

A function f that satisfies one of the conditions above is called **monotone**. If f satisfies (1) or (3), it is **monotone**, and if it satisfies (2) or (4), it is **strictly monotone**.

> **Remark**
>
> If f is differentiable, then:
>
> - $f' \geq 0$ implies f is **nondecreasing**.
> - $f' > 0$ implies f is **increasing**.
> - $f' \leq 0$ implies f is **nonincreasing**.
> - $f' < 0$ implies f is **decreasing**.

> **Remark**
>
> The sum of an increasing function and a decreasing function preserves the monotonicity properties.

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10. Injectivity and Monotonicity

We can check if f is **injective** by examining monotonicity. If f is **strictly monotonic**, then f is necessarily **injective** because:

- If $x_1 \neq x_2$, then either $x_1 > x_2$ or $x_2 > x_1$.
- If f is strictly monotonic, then either $f(x_1) < f(x_2)$ or $f(x_1) > f(x_2)$, implying $f(x_1) \neq f(x_2)$.

Thus, f is injective.

11. Inverse Function

Let $f : D \to R(f)$ be a bijective function. We define the inverse function g as follows:

$$g : R(f) \to D$$

such that

$$(g \circ f)(x) = x \quad \text{and} \quad (f \circ g)(x) = x$$

Thus, $g$is the **inverse function** of f and is unique.

Here is your content rewritten with the correct formatting for mathematical symbols using double dollar signs:

Inverse Image Properties

1. **Unions**:

   $$f^{-1}\left[\bigcup_{i \in I} U_i\right] = \bigcup_{i \in I} f^{-1}(U_i)$$

2. **Intersections**:

   $$f^{-1}\left[\bigcap_{i \in I} U_i\right] = \bigcap_{i \in I} f^{-1}(U_i)$$

3. **Complements**:

   $$f^{-1}(U^c) = (f^{-1}(U))^c$$

4. **Set Differences**:

   $$f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V)$$

5. **Symmetric Differences**:

   $$f^{-1}(U \Delta V) = f^{-1}(U) \Delta f^{-1}(V)$$

Miscellaneous Identities for Inverse Image:

1. $f|_X^{-1}(Y) = X \cap f^{-1}(Y)$
2. $f(f^{-1}(Y)) = Y \cap f(A)$
3. $X \subset f^{-1}(f(X))$, with equality if f is injective.

Direct Image Properties

Let $(f : A \to B)$ be a function and $(U \subset A)$.

1. **Unions**:

   $$f\left[\bigcup_{i \in I} U_i\right] = \bigcup_{i \in I} f(U_i)$$

2. **Intersections**:

   $$f\left[\bigcap_{i \in I} U_i\right] \subset \bigcap_{i \in I} f(U_i)$$

3. **Set Difference**:

   $f(V \setminus U) \supset f(V) \setminus f(U)$

   In particular, the complement of \(U \) satisfies:

   $f(U^c) \supset f(A) \setminus f(U)$

4. **Subsets**:

   If \(U \subset V \subset A \), then:

   $f(U) \subset f(V) \subset B$

5. **Inverse Image of a Direct Image**:

   $f^{-1}(f(U)) \supset U$

   with equality if (f) is injective.

6. **Direct Image of an Inverse Image**:

   $f\left[f^{-1}(V)\right] \subset V$

> **Remark**
>
> We search for g by solving the equation $y = f(x)$ and finding $x = g(y)$.
>
> If f is increasing and bijective on the interval $\left[\frac{\pi}{2}, -\frac{\pi}{2}\right]$, then f admits an inverse map denoted by $f^{-1} = \arcsin$, and we have:
>
> $\arcsin(\sin(x)) = x \quad \text{for} \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
>
> The inverse is defined by:
>
> $\sin(\arcsin(x)) = x \quad \text{for} \quad -1 \leq x \leq 1$

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12. Inverse Image

Let $f : D \to \mathbb{R}$ be a function that is not necessarily bijective. We define the **inverse image** of a set $B \subset \mathbb{R}$ as:

$f^{-1}(B) = \{x \in D : f(x) \in B\}$

For example,

$f^{-1}(\{0,1,2,3\}) = \{0,1,-1,\sqrt{3},-\sqrt{3},-2,2\}$

Theorem on Pre-images of Unions and Intersections

Let f be a real function, $f : \mathbb{R} \to \mathbb{R}$, and let U be a subset of $\mathbb{R}$.

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Let $F(D,R)$ denote the set of all functions defined on D with their range included in R. The set $F(D,R)$ is a **vectorial space**.

Properties of Functions

1. A function f is said to be **nonnegative** if:

   $$\forall x \in D, f(x) \geq 0.$$

2. A function f is said to be **positive** if:

   $$\forall x \in D, f(x) > 0.$$

3. A function f is said to be **negative** if:

   $$\forall x \in D, f(x) < 0.$$

4. We say $f \leq g$ if:

   $$\forall x \in D, f(x) \leq g(x).$$

> **Remark**
>
> The space $F(D,R)$ is not totally ordered.

▼ Limits of Functions

Let f be a function defined on D, and let x_0 be an accumulation point of D (not necessarily belonging to D). We say that f converges to a real number l as $x \to x_0$ if:

Notation

- $\lim_{x \to x_0} f(x) = l$
  means the limit of
  $f(x)$ as $x \to x_0$ is l.
- $\lim_{x \to x_0^+} f(x) = l$
  represents the limit of
  $f(x)$ as x approaches x_0 from the right-hand side.
- Similarly, $\lim_{x \to x_0^-} f(x) = l$
  represents the left-hand limit.

Theorem: Uniqueness of Limits

If a limit exists, it is unique.

▼ Proof

Suppose $\lim_{x \to x_0} f(x) = l_1$ and $\lim_{x \to x_0} f(x) = l_2$. Then:

$0 < |l_1 - l_2| = |l_1 - f(x) + f(x) - l_2| \leq |f(x) - l_1| + |f(x) - l_2|$.

Given the definitions of limits, this leads to a contradiction unless $l_1 = l_2$.

2. Limits at Infinity

- $\lim_{x \to x_0} f(x) = +\infty$ if:

$$\forall B > 0, \exists \delta > 0 \text{ such that } 0 < |x - x_0| < \delta \implies f(x) > B.$$

- $\lim_{x \to x_0} f(x) = -\infty$ if:

$$\forall B > 0, \exists \delta > 0 \text{ such that } 0 < |x - x_0| < \delta \implies f(x) < -B.$$

- $\lim_{x \to -\infty} f(x) = l$ if:

$$\forall \epsilon > 0, \exists A > 0 \text{ such that } x < -A \implies |f(x) - l| < \epsilon.$$

- $\lim_{x \to +\infty} f(x) = l$ if:

$$\forall \epsilon > 0, \exists A > 0 \text{ such that } x > A \implies |f(x) - l| < \epsilon.$$

- $\lim_{x \to +\infty} f(x) = +\infty$ if:

$$\forall B > 0, \exists A > 0 \text{ such that } x > A \implies f(x) > B.$$

- $\lim_{x \to -\infty} f(x) = -\infty$ if:

$$\forall B > 0, \exists A > 0 \text{ such that } x < -A \implies f(x) < -B.$$

- $\lim_{x \to -\infty} f(x) = +\infty$ if and only if:

$$\forall B > 0, \exists A > 0 \text{ such that } \forall x \in \mathbb{R},\ x < -A \implies f(x) > B.$$

- $\lim_{x \to -\infty} f(x) = -\infty$ if and only if:

$$\forall B > 0, \exists A > 0 \text{ such that } \forall x \in \mathbb{R},\ x < -A \implies f(x) < -B.$$

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3. Main Theorems

Sequential Characterization of Limits

Let f be a function defined on D, and let x_0 be an accumulation point of D. Then the following are equivalent:

1. $\lim_{x \to x_0} f(x) = l$.
2. For all sequences $(x_n) \subset D$ such that $x_n \to x_0$, we have $f(x_n) \to l$.

▼ Proof

- **(1) ⇒ (2):**

  Suppose $\lim_{x \to x_0} f(x) = l$. Then, for any sequence (x_n) converging to x_0:

  $$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |x - x_0| \leq \delta \implies |f(x) - l| \leq \epsilon.$$

  Hence,
  $|f(x_n) - l| \leq \epsilon$ for sufficiently large n.

- **(2) ⇒ (1):**

  Suppose $\lim_{x_n \to x_0} f(x_n) = l$ for every sequence (x_n), but $\lim_{x \to x_0} f(x) \neq l$. Then there exists $\epsilon > 0$ such that:

  $$\forall \delta > 0, \exists x \text{ with } |x - x_0| \leq \delta \text{ and } |f(x) - l| > \epsilon.$$

  Construct a sequence
  (x_n) converging to x_0 that violates the sequential limit, leading to a contradiction.

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4. Limit Algebra Rules

Let f and g be functions defined on the same domain D, and let x_0 be an accumulation point. If:

$$\lim_{x \to x_0} f(x) = l_1 \quad \text{and} \quad \lim_{x \to x_0} g(x) = l_2,$$

then:

1. $\lim_{x \to x_0} (f(x) + g(x)) = l_1 + l_2$.
2. $\lim_{x \to x_0} (f(x) \cdot g(x)) = l_1 \cdot l_2$.
3. $\lim_{x \to x_0} (\alpha \cdot f(x)) = \alpha \cdot l_1 \quad (\alpha \in \mathbb{R})$.
4. $\lim_{x \to x_0} |f(x)| = |l_1|$.
5. $\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{l_1}{l_2}, \quad \text{if } l_2 \neq 0$.

proof

Like sequences

💡

Theorem

Let f and g be functions defined on D, and suppose:

$\forall x \in D, f(x) \leq g(x)$.
Then:

$\lim_{x \to x_0} f(x) \leq \lim_{x \to x_0} g(x)$.

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Corollary 4.2

Let $f : D \subset \mathbb{R} \to \mathbb{R}$ be a function, and let x_0 be an accumulation point of D. Suppose that $M_1 < f(x) < M_2$ for all $x \in D$, and suppose that:

$\lim_{x \to x_0} f(x) = L$

Then:

$M_1 \leq L \leq M_2$

Proof: احشم (This proof is considered trivial or straightforward.)

Corollary 4.3

Let $f : D \subset \mathbb{R} \to \mathbb{R}$ be a function, and let x_0 be an accumulation point of D. If:

$\lim_{x \to x_0} f(x) = l \in \mathbb{R}$

then:

$\exists \delta > 0 \, \text{such that for} \, 0 < |x - x_0| < \delta, \ f(x) \, \text{is bounded.}$

Proof:

By the definition of a limit, we have:

$\forall \epsilon > 0, \, \exists \delta > 0 \, \text{such that} \, 0 < |x - x_0| < \delta \implies |f(x) - l| < \epsilon$

Therefore:

$l - \epsilon \leq f(x) \leq l + \epsilon$

This implies that $f(x)$ is bounded within the interval $[x_0 - \delta, x_0 + \delta]$.

Theorem

Let $f, g : D \subset \mathbb{R} \to \mathbb{R}$ be functions, and let x_0 be an accumulation point of D. If:

- g is bounded on D, and
- $\lim_{x \to x_0} f(x) = 0$,

then:

$\lim_{x \to x_0} f(x)g(x) = 0$

Proof:

Using the **squeeze theorem**:

- Since $f(x) \to 0$ as $x \to x_0$, for any $\epsilon > 0$, there exists $\delta > 0$ such that:

$|f(x)| < \frac{\epsilon}{M}, \quad \forall x \text{ where } 0 < |x - x_0| < \delta$

where M is the bound of $|g(x)|$, i.e., $|g(x)| \leq M$ for all $x \in D$.

- Then:

$|f(x)g(x)| \leq |f(x)| \cdot M < \frac{\epsilon}{M} \cdot M = \epsilon$

Thus:

$\lim_{x \to x_0} f(x)g(x) = 0$

Corollary

Suppose that:

$\lim_{x \to x_0} f(x) = L, \quad \lim_{x \to x_0} g(x) = M$

and x_0 is a point of accumulation of $D_f \cap D_g$. Then:

$\lim_{x \to x_0} \max\{f(x), g(x)\} = \max\{L, M\}$

and:

$\lim_{x \to x_0} \min\{f(x), g(x)\} = \min\{L, M\}$

Proof:

The first result follows from the identity:

$\max\{f(x), g(x)\} = \frac{f(x)+g(x)}{2} + \frac{|f(x)-g(x)|}{2}$

By the theorem on limits of sums and limits of absolute values, we can take the limit as $x \to x_0$:

$\lim_{x\to x_0} \max\{f(x), g(x)\} = \frac{L+M}{2} + \frac{|L-M|}{2} = \max\{L, M\}$

Similarly, for the minimum:

$\min\{f(x), g(x)\} = \frac{f(x)+g(x)}{2} - \frac{|f(x)-g(x)|}{2}$

Taking the limit as $x \to x_0$:

$\lim_{x\to x_0} \min\{f(x), g(x)\} = \frac{L+M}{2} - \frac{|L-M|}{2} = \min\{L, M\}$

▼ Continuous Functions

Definition:

Let $f : D \to \mathbb{R}$, and let $x_0 \in D$. We say that f is continuous at $x_0$if:

$\forall \epsilon > 0, \exists \delta = \delta(\epsilon) > 0, \forall x \in D : (0 < |x - x_0| < \delta) \implies |f(x) - f(x_0)| < \epsilon$

We denote this by:

$\lim_{x\to x_0} f(x) = f(x_0)$

Continuity from the left:

A function f is said to be continuous from the left at x_0 if:

$\lim_{x\to x_0^-} f(x) = f(x_0)$

Continuity from the right:

A function f is said to be continuous from the right at x_0 if:

$\lim_{x\to x_0^+} f(x) = f(x_0)$

Example:

Consider the function:

$$f(x) = \begin{cases} \frac{\sin(x)}{x}, & x \neq 0 \\ b, & x = 0 \end{cases}$$

We calculate:

$\lim_{x\to 0^-} f(x) = 1, \quad \lim_{x\to 0^+} f(x) = 1$

If $b = 1$, then f is continuous at $x_0 = 0$.

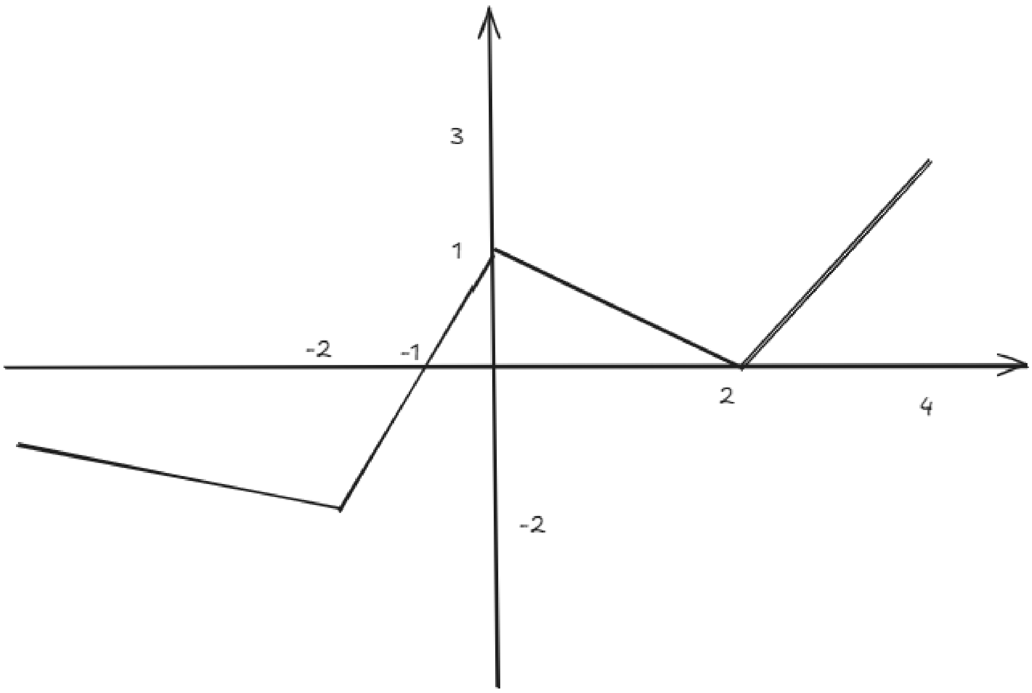
Definition for closed intervals:

A function f defined on a closed interval $[a, b]$ is said to be continuous:

- For all $x \in]a, b[$, it is continuous.
- At $x_0 = a$, it is right-continuous.
- At $x_0 = b$, it is left-continuous.

Example:

The function $f(x) = \sqrt{1 - x^2}$ is continuous on $[-1, 1]$.



▼ 1. Types of Discontinuities

Definition:

A function f is said to be discontinuous at x_0 if it is not continuous at x_0.

Types of Discontinuities:

1. **Removable Discontinuities:**

   This occurs when:

   $\lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x) = L$

but $f(x_0)$ is either undefined or not equal to L.

The discontinuity can be "removed" by redefining:

$$f(x) = \begin{cases} f(x), & x \neq x_0 \\ L, & x = x_0 \end{cases}$$

Example:

$$f(x) = \begin{cases} \frac{\sin(x)}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

Here, $\lim_{x \to 0^-} f(x) = 1$ and $\lim_{x \to 0^+} f(x) = 1$, so the discontinuity is removable.

2. **Discontinuity of the First Kind:**

   A function f has a discontinuity of the first kind at x_0 if:

   $$\lim_{x \to x_0^-} f(x) \text{ and } \lim_{x \to x_0^+} f(x) \text{ both exist but are not equal.}$$

   Example:

   $$f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 5, & x = 0 \end{cases}$$

3. **Discontinuity of the Second Kind:**

   A function f has a discontinuity of the second kind at x_0 if either:

   $$\lim_{x \to x_0^-} f(x) \text{ or } \lim_{x \to x_0^+} f(x) \text{ does not exist.}$$

   Example: Consider a step function that jumps infinitely near x_0.

Theorems

Theorem 1:

Let $f : D \to \mathbb{R}$ be a function, and let $x_0 \in D$. The following statements are equivalent:

1. f is continuous at x_0.
2. For any sequence $(x_n) \in D$ such that $x_n \to x_0$, we have:

   $f(x_n) \to f(x_0)$

▼ Proof:

Assume f is continuous at x_0. Take $l = f(x_0)$. Then, for all $\epsilon > 0$, there exists $\delta > 0$ such that:

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon$$

Thus, continuity implies the sequential property.

Theorem 2 (Algebra of Continuous Functions):

If f and g are continuous at x_0, then:

1. $f + g$ is continuous at x_0.
2. $f - g$ is continuous at x_0.
3. fg is continuous at x_0.
4. If $g(x_0) \neq 0$, then $\frac{f}{g}$ is continuous at x_0.

💡 **Theorem 3:**

If f is continuous on $[a, b]$, then f is bounded.

Proof (By Contradiction):

Assume f is not bounded on $[a, b]$. Then for all $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ such that:

$$f(x_n) \geq n$$

The sequence (x_n) is bounded, so by the Bolzano-Weierstrass theorem, there exists a subsequence (x_{n_k}) that converges to some $\xi \in [a, b]$.

By continuity:

$$f(x_{n_k}) \to f(\xi)$$

But this contradicts $f(x_{n_k}) \to +\infty$. Hence, f must be bounded.

💡 **Theorem: Boundedness and Attainment of Extrema**

If a function f is continuous on a closed interval $[a, b]$, then it attains its bounds at least once in $[a, b]$.

▼ **Proof**

1. **Case: Constant Function**

   If f is a constant function, then it attains its bounds at every point of the interval.

2. **Case: Non-Constant Function**

   Let f be a function that is not constant. Since f is continuous on the closed interval $[a, b]$, it is bounded.

   Let m and M be the infimum and supremum of f, respectively.

   We need to show that $\exists\, \alpha, \beta \in [a, b]$ such that:

   $$f(\alpha) = m \quad \text{and} \quad f(\beta) = M.$$

3. **Consider the Supremum M**

   Suppose f does not attain the supremum M, so that:

   $$f(x) \neq M \quad \forall x \in [a, b].$$

   Define the function:

   $$g(x) = \frac{1}{M - f(x)}, \quad \forall x \in [a, b].$$

   This function is positive for all $x \in [a, b]$.

4. **Properties of $g(x)$**

   Since f is continuous and $M - f(x) > 0$ for all x, $g(x)$ is also continuous and bounded on $[a, b]$.

   Let $k > 0$ be the supremum of $g(x)$. Then:

   $$\frac{1}{M - f(x)} \leq k, \quad \forall x \in [a, b].$$

   This implies:

$f(x) \leq M - \frac{1}{k}, \quad \forall x \in [a, b].$

This is a contradiction, as M was assumed to be the supremum. Hence, f must attain the value M.

By a similar argument, f attains the infimum m.

Theorem: Continuity and Local Sign Preservation

If a function f is continuous at an interior point c of an interval $[a, b]$ and $f(c) \neq 0$, then there exists a neighborhood N of c where $f(x)$ has the same sign as $f(c)$ for all $x \in N$

▼ Proof

By the definition of continuity, for any $\epsilon > 0$, there exists $\delta = \delta(\epsilon) > 0$ such that:

$$0 < |x - c| < \delta \implies |f(x) - f(c)| < \epsilon.$$

1. **Case:** $f(c) > 0$

   Choose $\epsilon = \frac{f(c)}{2} > 0$. Then for $x \in (c - \delta, c + \delta)$:

   $$f(c) - \epsilon < f(x) < f(c) + \epsilon.$$

   Substituting $\epsilon = \frac{f(c)}{2}$:

   $$\frac{f(c)}{2} < f(x) < \frac{3f(c)}{2}.$$

   Thus, $f(x) > 0$ in this neighborhood.

2. **Case:** $f(c) < 0$

   Choose $\epsilon = -\frac{f(c)}{3} > 0$. Then for $x \in (c - \delta, c + \delta)$:

   $$f(c) - \epsilon < f(x) < f(c) + \epsilon.$$

   Substituting $\epsilon = -\frac{f(c)}{3}$:

   $$\frac{2f(c)}{3} < f(x) < \frac{f(c)}{3}.$$

   Thus, $f(x) < 0$ in this neighborhood.

💡 **Theorem: Intermediate Value Theorem (IVT)**

Let f be a continuous function on $[a, b]$.

If $f(a) \cdot f(b) < 0$, then $\exists\, \alpha \in]a, b[\text{ such that } f(\alpha) = 0$.

▼ Proof

1. **Case:** $f(a) > 0$ **and** $f(b) < 0$

   Define the set:

   $$S = \{x \in [a, b] : f(x) > 0\}.$$

   - Since $f(a) > 0$, we have $a \in S$, so $S \neq \emptyset$.
   - The set S is bounded, as $S \subseteq [a, b]$.

2. **Existence of Supremum**

   By the completeness property of real numbers, the supremum of S exists. Let:

   $$\alpha = \sup S.$$

3. **Case:** $\alpha \neq a$

   Since $\alpha = \sup S$ and f is continuous at α:

   - For any $\gamma > 0$, $\forall x \in [a, a + \gamma]$, we have $f(x) > 0$.
   - This would imply $a + \gamma \in S$, contradicting $\alpha = \sup S$.
     Therefore,
     $\alpha \neq a$.

4. **Case:** $\alpha \neq b$

   Since $f(b) < 0$ and f is continuous at b:

   - There exists $\beta > 0$ such that $\forall x \in [b - \beta, b]$, $f(x) < 0$.
   - However, this would imply that $b - \beta$ is an upper bound of S, contradicting $\alpha = \sup S$.
     Therefore,
     $\alpha \neq b$.

Conclusion

Thus, $\alpha \in]a, b[$, and by the continuity of f at α:

$$f(\alpha) = 0.$$

Let $\alpha = \sup S$, where $S = \{x \in [a, b] : f(x) > 0\}$. The following properties hold:

1. $\forall x \in S : x \leq \alpha.$
2. $\forall \epsilon > 0, \exists x_\epsilon \in S : \alpha - \epsilon < x_\epsilon \leq \alpha.$
3. Specifically, for $\epsilon = \frac{1}{n}, \forall n \in \mathbb{N}:$
   - $n = 1 \implies x_1 \in S : \alpha - 1 < x_1 \leq \alpha,$
   - $n = 2 \implies x_2 \in S : \alpha - \frac{1}{2} < x_2 \leq \alpha,$
   - $n \implies x_n \in S : \alpha - \frac{1}{n} < x_n \leq \alpha.$

Proof by Contradiction

Let $x_n \in S$ be a sequence where $\lim_{n\to\infty} x_n = \alpha$.

Since $f(x) > 0$ for all $x \in S$ and f is continuous at α, we have:

$$\lim_{n\to\infty} f(x_n) = f(\alpha).$$

Thus, $f(\alpha) > 0.$

However:

1. $\alpha \in]a, b[$, and f is continuous at α.
2. By continuity, $\exists \delta > 0$ such that $\forall x \in [\alpha - \delta, \alpha + \delta], f(x) > 0.$
3. This implies $\alpha + \delta$ is an upper bound of S, contradicting $\alpha = \sup S$.

Therefore, $f(\alpha) = 0$ must hold.

2. Applications

1. Example: Function with Discontinuity

> Define the function:
>
> $$f(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}.$$
>
> This function is not continuous at $x = 0$. However, $f(x)$ demonstrates the idea of attaining bounds for continuous functions.

> **2. Example: Trigonometric Function with a Root**
>
> Define
>
> $$k(x) = \begin{cases} \frac{\sin(x-1)}{x-2}, & x \neq 2 \\ 0, & x = 2 \end{cases}.$$
>
> Here, $\exists\, \alpha \in \mathbb{R}$ such that $f(\alpha) = 0$. For example:
>
> $$k(1) = 0.$$

Corollary

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f assumes any value between its extrema.

Proof

1. Consider a subinterval $[\alpha, \beta] \subseteq [a, b]$.
    - Let $M = \sup_{\alpha \leq x \leq \beta} f(x)$.
    - Let $m = \inf_{\alpha \leq x \leq \beta} f(x)$.
2. By the Intermediate Value Theorem, f attains every value in $[m, M]$.

💡 **Theorem: Intermediate Value Theorem (Generalized Form)**

Let $f : [a, b] \to \mathbb{R}$ be a continuous function.

If $f(a) \neq f(b)$, then $\forall A \in [f(a), f(b)]$, $\exists c \in]a, b[$: $f(c) = A$.

Proof

1. Define the auxiliary function:

   $$\phi(x) = f(x) - A,$$

   where $\phi : [a, b] \to \mathbb{R}$.

2. **Properties of $\phi(x)$:**

   - ϕ is continuous because f is continuous.
   - At the endpoints:

     $$\phi(a) = f(a) - A, \quad \phi(b) = f(b) - A.$$

3. If $\phi(a) \cdot \phi(b) < 0$, by the Intermediate Value Theorem, there exists $c \in]a, b[$ such that:

   $$\phi(c) = 0.$$

4. Substituting $\phi(c) = 0$ into the definition of ϕ:

   $$f(c) - A = 0 \implies f(c) = A.$$

Thus, $\forall A \in [f(a), f(b)]$, there exists $c \in]a, b[$ such that $f(c) = A$.

3. Uniform Continuity

Let $f : D \to \mathbb{R}$ be a function. We say that f is **uniformly continuous** on D if:

$$\forall \epsilon > 0,\ \exists \delta = \delta(\epsilon) > 0 \text{ such that } \forall x, y \in D,$$

$$|x - y| \leq \delta \implies |f(x) - f(y)| \leq \epsilon.$$

Examples of Uniform Continuity

Example 1: Function $f(x) = x^2$

Let:

- $D_1 = [1, 17]$ (closed interval),
- $D_2 =]-5, 4[$ (open interval).

Uniform Continuity on D_1

The function $f(x)$ is **uniformly continuous** on D_1 if:

$\forall \epsilon > 0, \exists \gamma(\epsilon) > 0, \forall x, y \in D_1, |x - y| \leq \gamma \implies |f(x) - f(y)| \leq \epsilon.$

Let $\epsilon > 0$. We calculate:

$|f(x) - f(y)| = |x^2 - y^2| = |x - y||x + y|.$

On the interval D_1, the maximum value of $|x + y|$ occurs when $x, y = 17$
:

$|x + y| \leq 34.$

Thus:

$|f(x) - f(y)| \leq 34|x - y|.$

To ensure $|f(x) - f(y)| \leq \epsilon$, choose:

$\gamma = \frac{\epsilon}{34}.$

Hence, $f(x) = x^2$ is uniformly continuous on D_1.

Example 2: Function $f(x) = x^2$ on $\mathbb{R}$

For $D = \mathbb{R}$, the function $f(x) = x^2$ is **not uniformly continuous**. Proof by counterexample:

Let $\epsilon = 1$. For any $\delta > 0$, we can choose:

$x_n = n + \frac{1}{n}, \quad y_n = n - \frac{1}{n},$

where $n \in \mathbb{N}$.

Then:

1. $|x_n - y_n| = \frac{2}{n}$, which can be made smaller than any $\delta > 0$ for sufficiently large n.

As $n \to \infty$, $|f(x_n) - f(y_n)|$ becomes arbitrarily large, violating uniform continuity.

Thus, $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Example 3: Function $f(x) = \sqrt{x}$ on $[0, +\infty[$

The function $f(x) = \sqrt{x}$ is continuous on $[0, +\infty[$.

To verify **uniform continuity**, for any $\epsilon > 0$, let:

$|f(x) - f(y)| = |\sqrt{x} - \sqrt{y}|$.

By the Mean Value Theorem for square roots:

$|\sqrt{x} - \sqrt{y}| = \frac{|x-y|}{\sqrt{c}}$,

A function ρ_3 is **uniformly continuous** on a set if:

$$\forall \epsilon > 0, \exists \delta(\epsilon) > 0 : \forall x, y \in [0, +\infty[, |x - y| \leq \delta \implies |\sqrt{x} - \sqrt{y}| \leq \epsilon$$

Proof

1. Start with:

   $$|\sqrt{x} - \sqrt{y}| = \frac{|x-y|}{\sqrt{x}+\sqrt{y}}$$

2. For any $x, y \geq 0$:

   $$\frac{|x-y|}{\sqrt{x}+\sqrt{y}} \leq \frac{|x-y|}{2\sqrt{a}}$$
   where
   $a = \min(x, y)$ and $\sqrt{x} + \sqrt{y} \geq 2\sqrt{a}$

Thus, $|\sqrt{x} - \sqrt{y}| \leq \frac{|x-y|}{2\sqrt{a}}$, ensuring uniform continuity.

Function Analysis

Example: $f(x) = \frac{\sin x}{x},\ D = [0, 3]$

- Define:

$$\tilde{f}(x) = \begin{cases} \frac{\sin x}{x}, & x \in (0,3] \\ 1, & x = 0 \end{cases}$$

- Continuity:

  $\tilde{f}(x)$ is continuous on $[0,3]$ because the limit as $x \to 0$ matches the value at $x = 0$:

  $\lim_{x\to 0} \frac{\sin x}{x} = 1.$

Example: $f(x) = \sin(x^2),\ D = \mathbb{R}$

- **Not uniformly continuous**:
  - Sequence:

    $x_n = \sqrt{2n\pi},\ y_n = \sqrt{2n\pi} + \frac{1}{2}\sqrt{n\pi}.$
  - Difference:

    $|f(x_n) - f(y_n)| > \epsilon$ for arbitrarily small $|x_n - y_n|$.

💡 **Theorem: Uniform Continuity of Continuous Functions**

Let $f : [a, b] \to \mathbb{R}$ be a continuous function on the closed interval $[a, b]$. Then, f is uniformly continuous on $[a, b]$.

▼ Proof by Contradiction:

Assume f is **not uniformly continuous**. Then:

$$\exists \epsilon > 0, \forall \delta > 0, \exists x, y \in [a, b] : |x - y| \leq \delta \text{ and } |f(x) - f(y)| > \epsilon.$$

This means we can construct sequences $\{x_n\}$ and $\{y_n\}$ such that:

1. $x_n, y_n \in [a, b]$,
2. $|x_n - y_n| \leq \frac{1}{n}$,
3. $|f(x_n) - f(y_n)| > \epsilon$.

Compactness of $[a, b]$:

- Since $[a, b]$ is compact, the sequences $\{x_n\}$ and $\{y_n\}$ are bounded.
- By the Bolzano-Weierstrass theorem, there exist subsequences $\{x_{n_k}\}$ and $\{y_{n_k}\}$ that converge to some point $c \in [a, b]$:

$$x_{n_k} \to c, \; y_{n_k} \to c.$$

such $x_{n_k} \to \xi$, $y_{n_k} \to \eta$ $\}$ $(x_{n_k}), (y_{n_k}) \in [a, b]$ (closed)

so $\xi, \eta \in [a, b]$. there for $\exists \varepsilon > 0$ st: $\forall n \in \mathbb{N}$:

$\exists (x_{n_k}), (y_{n_k}) \in [a, b]$ st $|x_{n_k} - y_{n_k}| \leq \frac{1}{k} \to 0$

$\Rightarrow (\xi = \eta)$ and $|f(x_{n_k}) - f(y_{n_k})| \to |f(\xi) - f(\eta)| \geq \varepsilon > 0$

contradiction with the continuity of f at $\xi = \eta$

~~with~~ **theorem** Let $f, g : [a, b] \to \mathbb{R}$ be two uniformly continuous functions on $[a, b]$

then

1- $f \pm g$ is uni. con on $[a, b]$

2- fg " " "

3- $\frac{f}{g}$ " " " $(g(x) \neq 0)$

4 $f \circ g$ " " when applicable.

$\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{k}; \forall x, y \in D: |x - y| \leq \delta$

$\Rightarrow |f(x) - f(y)| \leq k \cdot |x - y|$

$\leq k \cdot \delta$

$\leq \varepsilon$

Inverse theorem on Yt.

check.